higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CONSTRUCTION PLANNING
NQF LEVEL 2

(12010042)

27 February 2019 (X-Paper)
09:00–12:00

Calculators may be used.

This question paper consists of 8 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. The drawings, sketches and diagrams must be done in pencil.
5. The sketches and or diagrams must be neat, reasonably large, in proportion and fully labelled.
6. ALL the abbreviations and symbols must comply with the latest National Building Regulations and ALL relevant SANS (SABS) codes.
7. Write neatly and legibly.

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QUESTION 1: TERMINOLOGY AND COMPONENTS OF THE CONSTRUCTION INDUSTRY

1.1 It is important to know the technical language and correct construction terminology representing the items and structure of their trade.

Give ONE word/term for each of the following descriptions by choosing a word/ term from the list below. Write the answer next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

deviation; renovations; graphometer; curing; redundant; viaduct; gable; ratio

1.1.1 A graduated circle used to measure horizontal circles

1.1.2 Amount by which a measurement differs from what is specified

1.1.3 Process by which concrete hardens

1.1.4 Proportion or quantities of materials for a mixture

1.1.5 Repairs to a building so that it is in a good condition

(5 × 1) (5)

1.2 Give TWO reasons why national building regulations should be adhered to when building. (2)

[7]

QUESTION 2: MEASURING, SETTING OUT AND LEVELLING

2.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (2.1.1–.1.5) in the ANSWER BOOK.

2.1.1 A measuring wheel can be used to measure the circumference

2.1.2 In a right angle triangle if the smallest angle is Ø then
$\tan Ø = \frac{\text{opposite}}{\text{hypotenuse}}$

2.1.3 For a house to be perfectly square, the diagonal length across the area of a square house should be the same.

2.1.4 A two peg method is used to find the level difference of two places.

2.1.5 A ranging rod is used for taking readings while levelling.

(5 × 1) (5)

2.2 Explain how you would use a levelling instrument to calculate horizontal distance from the instrument to another point using the stadia hairs. (4)

2.3 You want to build a hexagonal house with sides of 2 m.

Explain how you would set out this on the ground. Use rough sketches to illustrate. (6)

2.4 Give FOUR disadvantages of using a water level for levelling. (4)

2.5 A reading of 950 is taken at a point B using a level. Then a reading to the benchmark 1 300 is taken with the level in the same position.

What will the elevation be of point B given the elevation of the benchmark is 100 meters above sea level? (4)

2.6 FIGURE 1 below shows a Vernier calliper which is used to obtain dimensions of objects too small to measure accurately.

Name the SIX parts of the Vernier calliper marked from 1–6 respectively.

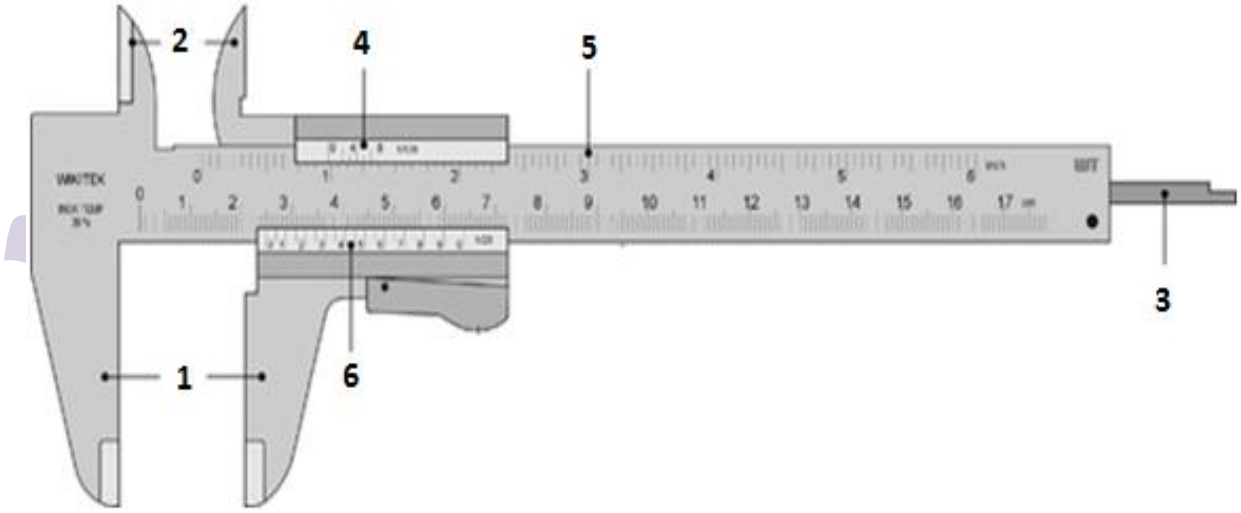


FIGURE 1 (6)

[29]

QUESTION 3: FOUNDATIONS

3.1 Explain why a site soil investigation is recommended before building a new structure. (3)

3.2 Explain the cases where raft foundations become the best option. (4)

3.3 What are the functions of a foundation? (3)

[10]

QUESTION 4: BEHAVIOUR OF STRUCTURES

4.1 Give ONE word/term for each of the following descriptions by choosing a word/term from the list below. Write only the word/term next to the question number (4.1.1–4.1.5) in the ANSWER BOOK.

short column; tension; torque; long column shear force; cantilever; springer; key stone; extrados; bulking

4.1.1 The voussoir located at the crown of the arch

4.1.2 A force that causes rotation

4.1.3 A force that tends to cut a component

4.1.4 The beam that is held and supported at one end only

4.1.5 A column with a slenderness ratio of less than 10

(5 × 1) (5)

4.2 Explain the following building terms:

4.2.1 Structural element (2)

4.2.2 Reinforcement (1)

4.3 FIGURE 2 below shows a vertical section through one side of a cantilever beam fixed wall. Reproduce the drawing in your ANSWER BOOK and indicate the area where tension and compression is located.

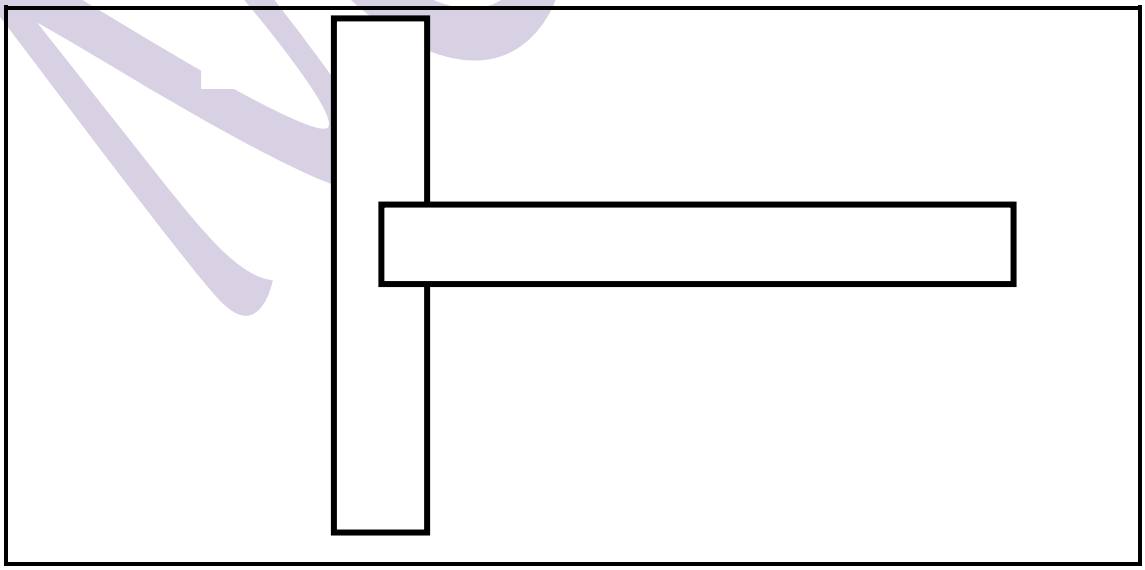


FIGURE 2 (2)

4.4 FIGURE 3 below shows an illustration of a frame structure commonly used in building construction and civil engineering.

Write the alphabet of the member AC, BC, CD and DA in the ANSWER BOOK and say whether the member is a 'Tie' or a 'Strut'.

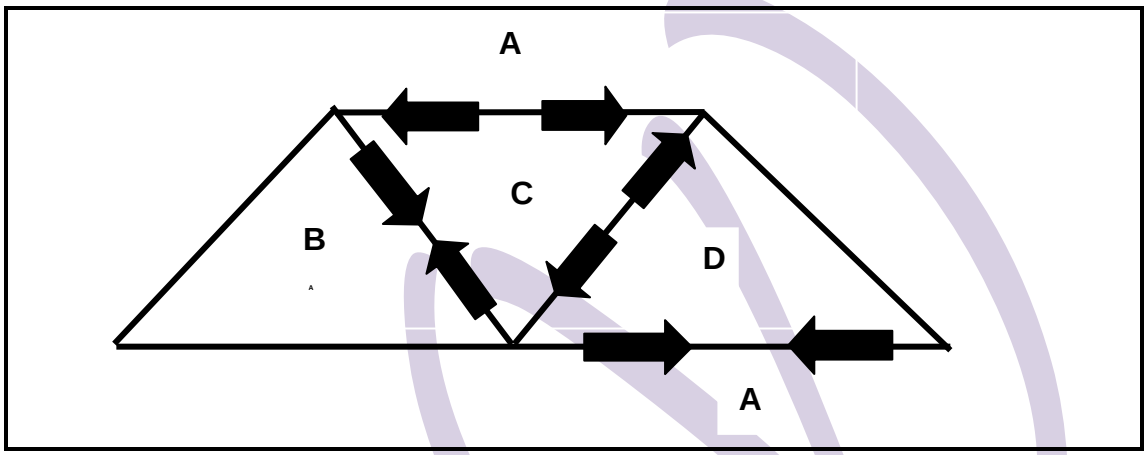


FIGURE 3 (4 × ½) (2)

4.5 Explain the difference between a *point load* and *uniform distributed load*. (2)

4.5 How would one differentiate between plane frames and space frames? (2)

4.7 Name the TWO kinds of loads that cause a five hinge and a four hinge failure of arches. (2)

[18]

QUESTION 5: INTERNATIONAL SYSTEM (SI) UNITS OF MEASUREMENTS

5.1 Convert the following:

5.1.1 .0000000007 to scientific notation (to the power of 10)

5.1.2 160 km to millimetres

5.1.3 2hr 25 min to seconds

5.1.4 25 mm^3 to m^3

5.1.5 Add (53° 48' + 27° 26')

(5 × 1) (5)

5.2 What are the units of measurements of the following:

5.2.1 Force

5.2.2 Pressure of a car's tyre

5.2.3 Current

5.2.4 Energy

5.2.5 Power

(5 × 1) (5)
[10]

QUESTION 6: AREAS AND VOLUMES

The following specifications are taken from a plan:

Length of the room 8 m
Width of the room 4,5 m
Height of the room 3,2 m
4 windows 2 m × 2 m each
Door 2 m × 2,8 m
1 litre of paint covers 4 m²

6.1 Calculate the concrete needed for a strip footing 600 mm × 200 mm all around. (3)

6.2 Calculate the total area of brick work. (5)

6.3 Calculate the litres of paint needed to paint both inside and outside. (2)
[10]

QUESTION 7: TECHNICAL DRAWING

7.1 Draw the hatching used for the following:

7.1.1 Brick

7.1.2 Stone

7.1.3 Insulation

7.1.4 Hard core

7.1.5 Screed

(5 × 1) (5)

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7.2 What do the following symbols or abbreviations on a building plan stand for?

7.2.1 RWP

7.2.2 BIC

7.2.3 SP

7.2.4 HR

7.2.5 SWP

7.2.6 FFL

(6 × 1) (6)

7.3 What are the following drawings used for:

7.3.1 Technical drawings

7.3.2 Engineering drawings

7.3.3 Service drawings

(3 × 1) (3)

7.4 Name the TWO drawings on a building plan needed to set out a new building. (2)

[16]

TOTAL: 100

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